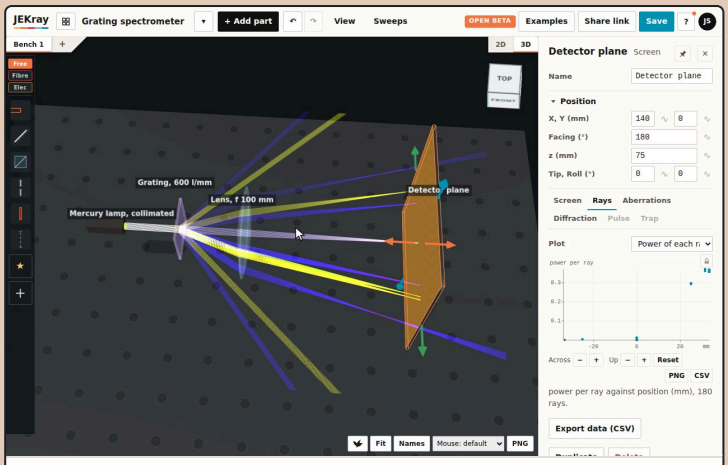
OPTICAL BENCH SIMULATOR · FREE IN YOUR BROWSER

JEKray

An optical bench simulator that shows the waves as well as the rays.

Lay out sources, lenses, mirrors, fibre and detectors on a virtual breadboard and see the light traced as you move them: exact rays through real glass, diffracted Gaussian beams with interference and polarisation, in 2D and 3D. Nothing to install, and it starts without an account.



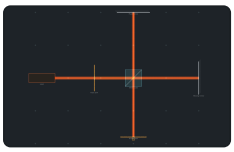
A grating spectrometer, turned in the 3D view. Drag any part and the light follows.

Checked against published results

Makers' lens files
Thorlabs and Edmund .zmx and .zar files import and focus to the makers' stated focal lengths.

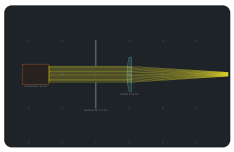
Published results
Fox and Li's resonator losses, Safronova's polarisabilities, measured rubidium absorption, a levitated nanosphere's damping.

Optical design
Real rays, Seidel aberrations and Strehl ratios of a singlet and a Cooke triplet against an independent ray tracer.



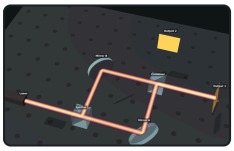
Interference, cavities and polarisation

Each ray carries its polarisation and optical path, so interferometers show fringes and cavities resonate. The diffracted field is worked out where rays cannot tell the story: slits, masks, a tight focus, a vortex beam.



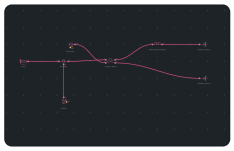
The makers' catalogues, built in

Schott, Ohara, Hoya, CDGM and Sumita, some 1,200 glasses, plus crystals and optical plastics. Import a Thorlabs or Edmund lens file; export a prescription your lens-design program opens.



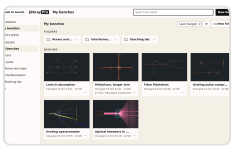
The diffracted field in 3D

Raise, tip and roll any part and the light follows. The field glows along each beam in calibrated W/cm², interference and all. Bring mounts and housings in from CAD as STL files.



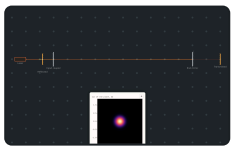
Fibre, free space and the signals between

Fibre subsystems, free-space optics and the electronics after the detector on one bench: photodiode signals, oscilloscopes, lock-ins and lock error signals. Pulses, nonlinear optics, vapour cells and optical traps too.



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48 examples and a two-minute tour

A Michelson, a Fabry–Pérot, a grating spectrometer, optical tweezers, a PDH lock, a mode converter: open one, break it, read the results. Documentation sets out every model with its equations.

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